

Intro

With over 240K records with 155 unique features, the main struggle was preparing the data for visualization and target variable creation. An initial inspection of the data and column name was completed. AI usage was to assist in coding. These sections were written in the notebook and the reasoning why. One note regarding AI was that, due to the complexity of this project, it often failed to generate solid advice. This showcased the limitations of AI and how additional time is needed to analyze and readjust AI's current logic compared to finding documents containing similar information.

Missing Values

Upon initial inspection of my missing values, a large proportion of features had missing values. I created a diagram showing features with missing values exceeding 10%. This is a strict threshold given the large number of features to evaluate. At that moment, I realized that my feature's had duplicate columns (e.g., "weathercondition" and "weatherconditionscode"). Using the provided Excel file column dictionary and my initial investigation, I removed roughly 80 features due to high missing values and column redundancy.

Manual Value Adjustment

Then I analyzed the remaining features using ".value_counts" to determine what needed adjustment. After adjusting, I manually removed the remaining undetermined missing values, resulting in roughly 137K records and 74 features.

Feature Separation

For efficiency testing, I asked AI to separate the remaining features into two categories: features used for modeling (Group A) and features used for target creation (Group B). This was another moment when the AI answer did not make sense (e.g., placing 'positivealcoholtest' in the modeling group, when this can only be examined after investigation). Thus, I had to manually group features afterward. "trainspeed" and "recordedestimatedspeed" were kept in the modeling group. Based on tonight's class discussion and continued progress on model building. I may examine if removing those two features makes more sense.

Target Variable Percent Calculation

Using Group B numeric features, a dictionary was created, with manual weight values assigned to each feature. To calculate the risk score, I first calculated the collective z-score and converted it to a percentage for distribution. Upon evaluation, large outlier values skewed the data sharply, producing an unrealistic percentage. Log transformation of $\log(1+x)$ was tested and saw no major improvement.

After segmenting the risk percentage to reveal the outliers, no obvious solution was present. AI was used to find another solution, which suggested an IQR threshold. By calculating my IQR score, the threshold was set to the value of " $Q3 + 1.5 * IQR$." Values exceeding it automatically placed at 100% as these are extraordinary/ catastrophic events skewing the data to the right. Results looked promising, but a large portion of the data was automatically set to 100, which could lead to poor modeling. Therefore, I manually set a threshold of 0.95 to restore a usable bell-shaped distribution of the risk percentage. The risk percentage variable was added to the Group A dataframe afterward.

Target Variable Category Calculation

Like the risk percentage, I assigned weights to the dictionary and separated them into financial and legal groups. The risk score per group was calculated using the same logic as risk percentage. However, upon writing this review, they were not logged and kept in their original collective z-score format. Financial risk had a solid bell-shaped distribution overall, while legal was extremely right skewed. Log transformation could improve the overall shape and will be tested during modeling to determine if it improves performance.

The classification method was broken down into the following labels: “Minimal Risk,” “Legal Risk,” Financial Risk,” and “Combination Risk.” The presence of the label was based on whether their collective z-score was greater than the mean (0). From that method, most of the data fell into “Minimal Risk,” followed by “Financial Risk” and “Combination Risk.” “Legal Risk” is the lowest at 8,078 records out of 130K. Therefore, in the next stage of this project, I will need to handle the class imbalance. The risk category variable was added to the Group A dataframe afterward.

Exploratory Data Analysis

EDA was conducted to observe how the data behaved after data cleaning and feature engineering. From the boxplot, there is evidence of category separation, with outliers present in the “Financial Risk” and “Minimal Risk” categories. The correlation matrix showed no strong correlations to the target variable “risk_percentage” with the highest being 0.35 for the train's “maximumspeed.” The heat map showed the following correlation between features: “trainspeed” + “maximumspeed” at 0.98, and “grossweight” + “loadedfreightcars” at 0.76.

One Hot Encoding/Separating Data/ Train-Test Split

After EDA, I one-hot encoded my categorical variables to prepare them for modeling, then split the dataframe into two copies for predicting “risk_percentage” and “risk_category.” The train-test split was performed only on the “risk_percentage” dataframe for learning purposes. Followed by using frequency counting for the remaining categorical variables to transform them into a percentage of frequency due to their high cardinality. This will also lower the dimensionality as having 50 columns representing each state in the one hot encoding would not be beneficial. This step had to be completed afterward due to possible data leakage. Finally, standard scaling was applied to the train-test split dataframe to reduce the influence of large values for linear regression and KNN modeling.